Manō : A Synthetic-Data-Driven Generative AI Framework for Privacy-Preserving, Proactive Mental Health Management

G. I. D. Weerasinghe
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22890382@my.sliit.lk

K. K. D. D. Keerthi
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22327208@my.sliit.lk

C. I. Wickramasuriya
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22890528@my.sliit.lk

D. G. M. Shalinda
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
it22317308@my.sliit.lk

K. T. S. Kasthuriarachchi
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
sanvitha.k@sliit.lk

U. M. M. P. K. Nawarathne
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
malithi.n@sliit.lk

***Abstract*—Mental health AI systems face a critical tension: real patient data, constrained by HIPAA and GDPR regulations, remains inaccessible for large-scale model development, while conventional synthetic data methods fail to capture the longitudinal, multimodal complexity of mental health trajectories. This paper presents Manō, a closed-loop generative AI framework operating under a zero-real-data paradigm that resolves this challenge through privacy-by-design architecture. The system comprises four tightly integrated subsystems orchestrated via Kafka event streaming. First, a synthetic data generation engine employs Conditional Tabular GAN (CTGAN) to produce demographically diverse patient profiles and TimeGAN to synthesize longitudinal wearable timeseries, achieving 87.49% statistical fidelity and 96% temporal autocorrelation preservation. Second, an intervention simulation module combines a Seq2Seq world model with Bahdanau attention (MSE: $1.41\text{x}10^{-4}$) for outcome forecasting and a Proximal Policy Optimization (PPO) agent for personalized treatment policy learning, enabling closed-loop refinement without real patient exposure. Third, a dual-branch Hybrid LSTM trained exclusively on synthetic data performs real-time cognitive risk prediction, delivering 96.2% classification accuracy, 0.98 high-risk recall, and 14ms inference latency. Fourth, a conversational support system integrates a fine-tuned Transformer LLM with a sentiment-aware empathy engine for risk-contextualized user engagement, while a Gaussian Mixture Model (GMM) with Expectation-Maximization enables dynamic peer clustering and collaborative filtering-based activity recommendations. All components operate on consumer-grade hardware (4GB VRAM) via mixed-precision training strategies. Experimental results demonstrate that this zero-real-data architecture achieves clinically plausible predictive accuracy and generative fidelity while maintaining strict privacy guarantees by construction, eliminating regulatory barriers inherent to real-data approaches.**

***Keywords*—synthetic data generation, TimeGAN, CTGAN, Seq2Seq world model, proximal policy optimization, LSTM, privacy-preserving machine learning, Gaussian mixture model**

I. INTRODUCTION

Mental health conditions affect over one billion individuals globally, yet computational approaches to screening, prediction, and intervention remain critically underdeveloped relative to other medical domains [1]. The primary barrier is not algorithmic but data-centric: real patient mental health data is protected under HIPAA, GDPR, and equivalent regulatory frameworks worldwide, requiring 6-24 months of institutional review board (IRB) approval for collection and imposing severe restrictions on cross-institutional sharing [2]. Publicly available mental health datasets contain fewer than 10,000 longitudinal patient records with multimodal sensor integration, insufficient for training deep learning systems that require diverse, high-dimensional training distributions [3].

Traditional approaches to synthetic health data generation via standard Generative Adversarial Networks suffer from mode collapse when applied to mixed-type tabular distributions, producing continuous approximations of discrete fields and destroying inter-feature correlations critical to clinical validity [4]. Time-series generation presents additional challenges: preserving autocorrelation structure, circadian physiological patterns, and cross-channel temporal dependencies requires architectural innovations beyond standard adversarial training [5].

This paper presents Manō, a closed-loop generative AI framework for personalized mental health support comprising four integrated components: (1) a synthetic data generation and intervention simulation engine implementing CTGAN, TimeGAN, Seq2Seq world modeling, and PPO reinforcement learning; (2) a cognitive risk prediction system utilizing hybrid LSTM with attention mechanisms for real-time multimodal classification; (3) a multi-persona conversational support system employing fine-tuned Transformer architecture with empathy-optimized response generation; and (4) a community-driven clustering and

activity recommendation system based on Gaussian Mixture Models with collaborative filtering.

The framework operates under a strict zero-real-data paradigm, achieving HIPAA/GDPR compliance by construction through 100% synthetic data pipelines. All components are designed for deployment on consumer-grade hardware (NVIDIA RTX 3050 Ti, 4GB VRAM) through mixed-precision training, gradient checkpointing, and memory-efficient architectural decisions. The system employs event-driven inter-component communication via Apache Kafka, enabling closed-loop feedback where downstream performance metrics refine upstream generation parameters.

The principal contributions of this work are: (i) a two-stage synthetic patient generation pipeline combining CTGAN for static demographics with TimeGAN for longitudinal wearable timeseries, achieving 87.49% statistical similarity and 96% autocorrelation preservation; (ii) a hybrid dual-branch LSTM architecture for multimodal risk classification achieving 96.2% accuracy with 14ms inference latency; (iii) a differentiable Seq2Seq world model enabling reinforcement learning-based treatment policy optimization via PPO with dual-head actor architecture; (iv) a privacy-preserving closed-loop architecture integrating synthetic generation, prediction, conversational support, and community clustering on consumer hardware; and (v) comprehensive empirical validation demonstrating clinically plausible performance across all subsystems.

II. LITERATURE REVIEW

Synthetic data generation for healthcare has evolved from simple perturbation methods to deep generative models. Choi et al. [6]. introduced medGAN for generating discrete patient records using autoencoders combined with GANs, though the approach was limited to binary and count features. Xu et al. [7] proposed CTGAN at NeurIPS 2019, introducing mode-specific normalization via Variational Gaussian Mixtures and conditional generation to address the mixed-type distribution problem in tabular data. TVAE [7]. offered a variational autoencoder alternative but produced blurred outputs incapable of capturing sharp distributional boundaries. Yoon et al. [8]. introduced TimeGAN, combining unsupervised GAN training with supervised temporal coherence loss to preserve autoregressive structure in synthetic timeseries, demonstrating superiority over recurrent conditional GAN (RCGAN) and Conditional GAN for timeseries on medical sensor data [9], [10].

Risk prediction in mental health has leveraged both classical machine learning and deep learning approaches. Reece and Danforth [11]. demonstrated that Instagram activity predicted depression with 70% accuracy using ensemble methods. Shen et al. [12] applied LSTM networks to electronic health records for suicide risk prediction, achieving AUC-ROC of 0.84. Hybrid architectures combining temporal and static branches have emerged as effective strategies for multimodal health data [13], [14]. The attention mechanism [15] provides interpretable temporal weighting, enabling clinicians to understand which historical events drive predictions.

Reinforcement learning for treatment optimization represents a growing domain. Gottesman et al. [16] applied off-policy RL to optimize sepsis treatment using observational clinical data. Wang et al. [17] demonstrated that PPO [18] with shaped reward functions could learn clinically plausible medication dosing policies. The challenge of learning treatment policies without real clinical trial data motivates the model distillation approach adopted in this work, where medical knowledge is encoded as mathematical rules, synthesized into training trajectories, and distilled into differentiable neural networks [19].

Conversational AI for mental health has progressed from rule-based systems like Woebot [20] to neural approaches. Rashkin et al. [21] introduced the EmpatheticDialogues dataset, enabling fine-tuning of empathetic response generation. Roller et al. [22] demonstrated that large-scale Transformer fine-tuning could produce contextually appropriate therapeutic responses. Lin et al. [23] combined sentiment analysis with BERT-based classifiers for real-time crisis detection in conversational contexts. Privacy-preserving training approaches using synthetic conversation data [24] and differential privacy [25] address the ethical constraints of mental health dialogue systems.

Community-based mental health interventions leverage clustering and recommendation techniques. Gaussian Mixture Models [26] provide probabilistic soft-clustering enabling multi-community participation, superior to hard-assignment methods like K-Means for modeling multidimensional resilience profiles [27]. Collaborative filtering and content-based recommendation [28] ,[29]. have been applied to wellness activity suggestion. The integration of clustering with recommendation within a closed-loop architecture that feeds into conversational support represents a novel contribution of this work.

The intersection of privacy-preserving computation and mental health AI has received increasing attention. Beaulieu-Jones et al. [60] demonstrated that GANs can generate synthetic clinical records passing statistical indistinguishability tests while preserving differential privacy guarantees. McMahan et al. [71] introduced federated averaging for distributed model training without centralizing sensitive data, and Li et al. [72]. surveyed emerging challenges in federated learning for healthcare applications. Dwork [79] formalized differential privacy as a mathematical framework applicable to neural network training through gradient perturbation [25]. However, the complete elimination of real data from the training pipeline, as implemented in the proposed framework, represents a more conservative privacy posture than differential privacy alone, eliminating both direct and inferential disclosure risks.

Uncertainty quantification in clinical AI systems remains an open challenge. Gal and Ghahramani [73] proposed Monte Carlo dropout as a Bayesian approximation for estimating model uncertainty, while Kendall and Gal [74] distinguished between aleatoric and epistemic uncertainty in deep learning for safety-critical applications. Hardware-efficient deep learning has emerged as a critical enabler: mixed precision training [46] reduces memory consumption via FP16 forward passes, and gradient checkpointing [47] trades computation for memory by recomputing activations during

backpropagation, collectively enabling complex multi-network architectures within consumer GPU constraints.

Mental health assessment instruments provide the clinical grounding for computational approaches. The DASS-21 scale [85] measures depression, anxiety, and stress across three subscales, while the Hamilton Depression Rating Scale [84] remains the gold standard for severity assessment. Computational approaches leveraging passive sensing data from smartphones [83], [88] and wearables [89] have demonstrated that longitudinal behavioral features predict mental health outcomes with clinically useful accuracy, motivating the multimodal sensor fusion approach adopted in this work.

III. METHODOLOGY

The Manō framework operates as a unified closed-loop architecture where four components exchange data through event-driven pipelines. Component 1 generates synthetic training data consumed by all downstream components. Component 2 produces real-time risk scores disseminated to Components 3 and 4 via synchronous API and asynchronous Kafka event streams. Component 3 consumes risk context for proactive dialogue initiation and emits engagement metrics to Component 4. Component 4 provides community insights back to Component 3 and receives aggregated feedback for upstream generation refinement. All inter-component communication operates on anonymized feature vectors with TLS 1.3 encryption and mutual TLS for internal gRPC calls.

A. Synthetic Data Generation and Adaptive Multimodal Intervention Simulation Engine

The synthetic data generation pipeline implements a two-stage architecture addressing the static-to-dynamic data challenge. Stage 1 employs CTGAN for static demographic generation. For each continuous column, CTGAN fits a Variational Gaussian Mixture (VGM) with k components, learning per-mode parameters:

$$p(x) = \Sigma(i = 1 \; to \; k) \, \pi_i \cdot N(x \mid \mu_i, \sigma_{i^2}) \quad (1)$$

where π_i denotes mixture weights, μ_i and σ_i represent mode-specific means and standard deviations. During generation, the model samples mode m ~ Categorical(π), then x ~ N(μ_m, σ_m), enabling faithful reproduction of multimodal distributions. The conditional generator accepts category embeddings c concatenated with noise vector z:

$$G(z, c) = FC^3\left(ReLU\left(FC^2\left(ReLU\left(FC^1([z;\, c])\right)\right)\right)\right) \quad (2)$$

where z ∈ R^{128} and c ∈ R^{64}. The discriminator employs separate sub-networks for categorical (cross-entropy) and continuous (L1 penalty) features with Wasserstein-style gradient penalty (λ=10). Training proceeds for 600 epochs with batch size 256, generator learning rate 2×10^{-4}, and 2:1 generator-to-discriminator update ratio. The model generates 10,000 synthetic profiles with 30 demographic features achieving 87.49% overall statistical similarity to reference distributions, with correlation preservation between 93-98% across feature pairs.

Stage 2 addresses temporal data generation through TimeGAN, comprising four interacting neural networks. The Embedder (GRU, 128 hidden units) compresses raw 7-day timeseries x ∈ R^{7x4} into latent representations h ∈ R^{7x128}. The Recovery network reconstructs features from latent space. The Supervisor enforces temporal coherence by predicting $h_{\{t+1\}}$ from h_t, ensuring the generator respects autoregressive structure. The Generator produces synthetic latent codes from noise z ~ N(0, I). Training proceeds in three phases:

Phase A (Epochs 1-50): Autoencoder pre-training minimizes reconstruction loss

$$L_R = \text{MAE}(\text{x}, \text{Recovery}(\text{Embedder}(\text{x}))) \quad (3)$$

achieving 0.045 loss. Phase B (Epochs 51-150): Supervisor training minimizes temporal prediction loss achieving 0.032 loss with frozen embedder. Phase C (Epochs 151-300): Joint adversarial training with composite generator loss:

$$L_G = 0.5 \cdot L_{adv} + 0.5 \cdot L_{supervisor} + 0.1 \cdot L_{moments} \quad (4)$$

where the moments matching loss prevents spectral collapse:

$$L_{moments} = \lambda^1 \cdot \|\mu_{real} - \mu_{fake}\| + \lambda^2 \cdot \|\sigma_{real} - \sigma_{fake}\| \quad (5)$$

with λ_1=0.05, λ_2=0.02. Convergence is achieved when generator and discriminator losses equilibrate at 0.5. The generated timeseries achieves 96% autocorrelation match, Inception Score at 98% of real data, and moment matching within 1% error.

A deterministic Medical Rule Engine produces ground truth labels. High Risk (2): stress > 0.7 AND sleep quality < 0.4 OR HR stability < 0.25. Medium Risk (1): stress > 0.6 OR sleep quality < 0.35. Low Risk (0): otherwise. The resulting class distribution (62:28:10 for Low: Medium: High) matches clinical epidemiology.

The intervention simulation subsystem implements a Seq2Seq LSTM with Bahdanau attention for outcome prediction. The encoder processes 7-day patient history through a 2-layer LSTM ($hidden_{\text{dim=256}}$). The attention mechanism computes context vectors:

$$\alpha_t = softmax(v^T \tanh(W[h_{dec};\, h_{enc}])) \quad (5)$$

$$c_t = \Sigma \, \alpha_t \cdot h_{enc[t]} \quad (6)$$

The decoder receives concatenated attention context (256-dim), intervention condition vector (6-dim: one-hot type + continuous intensity), and previous output (4-dim) at each timestep. Training on 50,000 virtual clinical trials with teacher forcing (ratio 0.5 decaying to 0.0) and mixed-

precision training achieves validation loss of 0.000139 (MSE).

Treatment optimization employs PPO with a dual-head actor network. The feature extractor processes state through Dense(256)→ReLU→Dense(128)→ReLU. The discrete actor head produces intervention type logits over 5 categories via Categorical distribution. The continuous actor head outputs intensity μ through sigmoid activation with learnable log-standard deviation:

$$\pi(a|s) = Cat(a_d \mid f_{d(s)}) \cdot N(a_c \mid \sigma(f_c(s)), \exp(\log_\sigma)) \quad (7)$$

The PPO objective with clipped surrogate:

$$L_{PPO} = -E[\min(r_t \cdot A_t, clip(r_t, 1-\varepsilon, 1+\varepsilon) \cdot A_t)] \quad (8)$$

where $r_t = \pi_{\theta(a_t|s_t)/\pi_\theta_old(a_t|s_t)}$, ε=0.2, and advantages A_t are computed via Generalized Advantage Estimation (γ=0.99). The shaped reward function $R = w_1 \cdot \Delta Risk - w_2 \cdot Intensity$ penalizes unnecessary treatment intensity. The critic loss $L_V = 0.5 \cdot (V(s)-R)^2$ and entropy bonus $H = -0.01 \cdot H(\pi)$ encourage exploration. Training over 5,000 episodes on RTX 3050 Ti converges to average reward 7.1±0.3, with the agent learning minimum-effective-dose behavior and patient-specific treatment matching.

B. Cognitive Risk Prediction

Component 2 implements hybrid dual-branch architecture for multimodal risk classification. Branch A (temporal) processes 7-day wearable sequences $x_{dyn} \in R^{7x4}$ through a 2-layer stacked LSTM with 128 hidden units and dropout 0.3:

$$h_t, c_t = LSTM(x_t, h_{\{t-1\}}, c_{\{t-1\}}); \; z_{temp} = h_T \in R^{128} \quad (9)$$

Branch B (static) processes 30-dimensional demographic features x_stat through Dense(128)→BatchNorm→ReLU→Dropout(0.3)→Dense(64)→BatchNorm→ReLU, producing z_stat ∈ R^{64}. The fusion layer concatenates both embeddings:

$$z_{fused} = [z_{temp}; \; z_{stat}] \in R^{192} \quad (10)$$

$$p(y|x) = softmax\left(Dense^3\left(ReLU\left(Dense^2\left(ReLU\left(Dense^1(z_{fused})\right)\right)\right)\right)\right) \quad (11)$$

Training employs weighted cross-entropy loss with class weights [1, 2.2, 6.2] addressing the 62:28:10 class imbalance. AdamW optimizer (lr=1×10^{-3}) with ReduceLROnPlateau scheduling (factor=0.5, patience=5) and early stopping (patience=20) ensures convergence without overfitting. The model achieves 96.2% overall accuracy, weighted precision 0.97, high-risk recall 0.98, weighted F1 0.97, and AUC-ROC 0.979 on held-out test data. Inference latency averages 14ms on CPU with model artifact size under 2MB, enabling edge deployment.

The trajectory forecasting engine enables proactive risk prediction through a hybrid mathematical-neural approach. Given a 3-day history buffer [T-2, T-1, T], linear trend detection computes feature slopes. A phantom day T+1 is simulated by linear extrapolation, normalized, and passed through the LSTM for forward-looking risk classification. This transforms the system from reactive classification to proactive alerting, detecting deteriorating trajectories before clinical threshold breach.

C. Multi-Persona Conversational Architecture

Component 3 implements an empathetic conversational support system comprising a fine-tuned Transformer LLM, sentiment analysis sidecar, empathy engine, and crisis detection pipeline. The base architecture employs a GPT-variant Transformer fine-tuned on synthetic mental health dialogue data generated by Component 1. The empathy engine applies post-processing incorporating CBT reframing rules, motivational interviewing patterns, and positive psychology templates to raw LLM outputs.

Sentiment analysis utilizes a BERT-based classifier operating as an asynchronous sidecar, classifying user messages along valence and arousal dimensions with specific attention to anxiety, depression, and crisis markers. Crisis detection implements keyword-based regex patterns operating independently of LLM inference, achieving recall exceeding 99% for crisis-indicative language with immediate escalation to safety protocols including validated resource links.

The system integrates real-time risk scores from Component 2 via REST API and Kafka event subscription. When risk levels elevate (e.g., transition from Medium to High), the proactive engagement engine triggers notification-based dialogue initiation. Session context is maintained in Redis with 30-minute TTL, supporting multi-turn conversations up to 50 turns. Token streaming via WebSocket delivers progressive response generation for perceived responsiveness, with first-token latency under 500ms and complete response within 2 seconds at P95.

The architecture employs event-driven microservice patterns with WebSocket gateway through NGINX. The frontend React SPA connects via WebSocket to the NLP inference service, which maintains session state and streams token-by-token responses. A separate sentiment analysis sidecar processes messages asynchronously, emitting events to the crisis detection pipeline and Component 4 engagement metrics stream. Conversation content is never persisted post-session, and aggregated metrics are transmitted with differential privacy guarantees.

D. Clustering and Recommendation System

Component 4 implements dynamic user clustering via Gaussian Mixture Models with activity recommendation through collaborative and content-based filtering. The resilience scoring service computes composite scores across four domains (physical, behavioral, emotional, social) from 15+ input features with normalized 0-100 scaling. The weighted composite score:

$$S_{overall} = 0.20 \cdot S_{physical} + 0.20 \cdot S_{behavioral} + 0.35 \cdot S_{emotional} + 0.25 \cdot S_{social} \quad (12)$$

The GMM clusters users into 5-15 dynamic groups with BIC/AIC-optimized component selection. For K mixture components with parameters $\Theta = \{\pi_k, \mu_k, \Sigma_k\}$:

$$p(x|\Theta) = \Sigma_{k=1}^{K} \pi_k \cdot N(x \mid \mu_k, \Sigma_k) \quad (13)$$

Parameter estimation via Expectation-Maximization alternates between computing posterior responsibilities (E-step) and updating parameters (M-step). The E-step computes:

$$\gamma_{nk} = \frac{\pi_k \cdot N(x_n \mid \mu_k, \Sigma_k)}{\Sigma_j \pi_j \cdot N(x_n \mid \mu_j, \Sigma_j)} \quad (14)$$

The M-step updates:

Mean update:

$$\mu_k = \frac{\Sigma_n \gamma_{nk} \cdot x_n}{\Sigma_n \gamma_{nk}}$$

Covariance update:

$$\Sigma_k = \frac{\Sigma_n \gamma_{nk} (x_n - \mu_k)(x_n - \mu_k)^T}{\Sigma_n \gamma_{nk}} \quad (15)$$

Optimal K is selected by minimizing BIC = $-2 \cdot \ln(L) + p \cdot \ln(n)$, where L is the maximized likelihood, p is the number of parameters, and n is the sample size. Incremental EM updates process new user data within 500ms without full model retraining, supporting 100K+ user profiles.

Probabilistic (soft) cluster memberships enable multi-community participation. A user with $\gamma = [0.72, 0.18, 0.10, ...]$ is primarily assigned to the dominant cluster while maintaining secondary affiliations. Peer support groups target optimal sizes of 8-12 members. The recommendation engine employs user-based and item-based collaborative filtering over a curated database of 21+ evidence-based wellness activities, with content-based matching between identified problem domains (scores below 40) and activity categories.

Cluster quality is monitored via silhouette coefficient and Davies-Bouldin index. Concept drift detection triggers model adaptation when cluster assignments change for more than 20% of users in a single cycle. Stability monitoring via Jensen-Shannon divergence (threshold > 0.1) ensures temporal consistency.

IV. RESULTS AND EVALUATION

Comprehensive evaluation spans all four components with quantitative metrics derived from held-out test sets, convergence analysis, and computational efficiency measurements. All experiments were conducted on NVIDIA RTX 3050 Ti (4GB VRAM) with PyTorch backend using mixed-precision training and gradient checkpointing.

CTGAN evaluation on 10,000 generated profiles demonstrates 87.49% overall statistical similarity via column-wise Kolmogorov-Smirnov tests. Feature-level analysis shows Age (98.8%), Work Years (97.5%), and Salary (96.3%) similarity. Correlation preservation ranges from 93% (Gender↔Role) to 96% (Age↔Work Experience, Remote↔Tech). Mode coverage analysis confirms capture of all distributional modes including multimodal features (e.g., trimodal age distribution with peaks at 25, 35, 50).

TimeGAN evaluation over 300 training epochs confirms convergence without oscillation. Reconstruction fidelity achieves MAE of 0.8 bpm for heart rate (2.3% error), 0.06 hours for sleep (0.9% error), 0.015 for stress (2.1% error), and 0.013 for quality (1.8% error). The embedder captures 99.1% of variance. Temporal consistency measured by autocorrelation matching achieves 96% for heart rate sequences. Latent space moment matching remains within 1% across all channels. Inception Score reaches 98% of real data baseline (8.21 vs. 8.34).

The Hybrid LSTM achieves 96.2% overall accuracy on the test set. Per-class performance: Low Risk (precision 0.935, recall 0.948, F1 0.941), Medium Risk (precision 0.872, recall 0.814, F1 0.842), High Risk (precision 0.824, recall 0.820, F1 0.822). AUC-ROC reaches 0.979 (one-vs-rest). The confusion matrix confirms zero High→Low misclassifications, validating safety-critical boundary separation. Comparative benchmarking against XGBoost (97.1% accuracy) reveals a 0.9% accuracy trade-off in exchange for temporal readiness and native forecasting capability.

The attention-enhanced LSTM from Component 2 with production configuration achieves 96.2% accuracy, high-risk precision 0.97, high-risk recall 0.98, and F1 0.97. Inference latency averages 14ms on CPU, supporting 70+ requests per second per node with P99 latency under 50ms. Model artifact size remains under 2MB, enabling mobile and edge deployment.

Seq2Seq validation demonstrates training loss convergence from 0.1247 to 0.000141 over 100 epochs, with validation loss tracking at 0.000139 confirming absence of overfitting. Attention analysis reveals progressive focus narrowing: uniform attention at epoch 1 evolves to sharp focus on days 6-7 by epoch 100, consistent with clinical expectations that recent physiological events are most predictive of treatment response.

PPO training over 5,000 episodes shows exploration phase (episodes 1-500, reward -2.0→2.5), learning phase (500-1500, reward 2.5→5.0), refinement phase (1500-3000, reward 5.0→6.5), and exploitation phase (3000-5000, reward 6.5→7.1±0.3). The converged policy demonstrates clinically interpretable behavior: selecting CBT for high-stress profiles, medication for severe cases, exercise/wellness for mild presentations, with intensity following minimum-effective-dose principles.

End-to-end system performance metrics: total pipeline latency 43ms, throughput 1,176 requests/second, total model size 118MB, active memory footprint 320MB. Component 4 achieves scoring latency under 100ms, cluster assignment under 200ms, and full analysis pipeline under 500ms per user.

Cross-component integration testing validates the closed-loop data flow. When Component 2 detects a risk transition (Medium→High) for a test profile, the Kafka event stream delivers the risk-score-event to Component 3 within 30 seconds, triggering proactive dialogue initiation. Component 3 subsequently invokes Component 1 intervention simulation

API, receiving treatment recommendations (e.g., CBT at intensity 0.72) with predicted risk reduction from 0.78 to 0.25 within 180ms. Component 4 consumes both risk events and conversation engagement metrics, producing updated cluster assignments and activity recommendations within the 15-minute cron cycle.

Memory profiling during TimeGAN joint training (the most demanding phase) confirms operation within 4GB VRAM: Embedder+Recovery consume 2.1GB, batch gradients 1.2GB, optimizer states 0.5GB, with gradient checkpointing maintaining total allocation at 3.95GB. Without gradient checkpointing, the joint phase would require 8.2GB, exceeding available VRAM. Mixed-precision training provides approximately 2x memory reduction and 1.5x speedup across all training phases.

V. DISCUSSION AND ETHICAL CONSIDERATIONS

The zero-real-data architecture eliminates the single largest risk vector in healthcare AI systems: patient data breach. By operating exclusively on synthetic data, the framework achieves HIPAA/GDPR compliance by construction, eliminates IRB delays, and enables unlimited training data diversity. The trade-off is potential synthetic-to-real distribution shift, where models trained on synthetic data may underperform on real-world edge cases including rare comorbidities and atypical physiological profiles.

The deterministic Medical Rule Engine introduces a known limitation: ground truth labels derive from heuristic rules rather than clinical diagnoses. While the 62:28:10 class distribution matches epidemiological expectations, boundary cases near decision thresholds may not reflect clinical ambiguity. Future integration of federated learning would enable the system to learn from real clinical data without centralizing it, addressing the synthetic-to-real gap while maintaining privacy guarantees.

The 0.9% accuracy trade-off between XGBoost (97.1%) and the Hybrid LSTM (96.2%) represents a deliberate architectural decision prioritizing temporal readiness over marginal static accuracy. The LSTM architecture enables native sequence processing for continuous wearable streams without requiring feature engineering or model retraining, whereas XGBoost would require complete architectural redesign for longitudinal monitoring.

Hardware-aware optimizations demonstrate that research-grade deep learning pipelines can operate on consumer hardware. Mixed-precision training (FP16) reduces memory consumption by approximately 2x, gradient checkpointing provides 3x memory reduction at 0.5x computational cost, and reduced batch sizes with gradient accumulation (8 steps of 32 = effective batch 256) maintain learning dynamics. The complete training pipeline fits within 3.95GB during joint TimeGAN training, the most memory-intensive phase.

Ethical considerations extend beyond privacy to therapeutic responsibility. The system provides non-diagnostic mental health guidance and maintains clear disclaimers about its non-clinical status. Crisis detection operates independently of LLM inference via regex-based keyword matching, ensuring safety-critical functionality persists even during model failures. The empathy engine applies evidence-based therapeutic frameworks (CBT, motivational interviewing) rather than unconstrained language generation, reducing the risk of harmful responses.

Scalability analysis identifies Component 3 (Transformer inference) as the primary bottleneck due to GPU-dependent inference. Mitigation strategies include response caching for frequently asked questions, model distillation for reduced compute footprint, and vLLM continuous batching for throughput optimization. Component 2 scales linearly with CPU replicas (70 RPS/node). Component 4 employs incremental EM updates avoiding O(n) recomputation, supporting 100K+ user profiles with sub-second assignment.

Limitations of this work include the absence of real clinical validation, the reliance on rule-based ground truth labels, the 7-day prediction horizon (insufficient for slow-acting treatments), point predictions without uncertainty quantification, and a single universal treatment policy rather than hierarchical subgroup-specific policies. Future work will address these through Bayesian neural networks for uncertainty estimation, 30-day prediction horizons, adverse effect modeling, hierarchical RL with subgroup-specific policies, and federated learning integration for privacy-preserving real data incorporation.

The inter-component feedback architecture warrants specific discussion. The closed-loop design enables Component 1 to refine generation parameters based on aggregated, anonymized conversation success metrics from Component 3, improving synthetic conversation data realism over iterative cycles. Similarly, Component 4 cluster quality metrics inform Component 1 diversity requirements, ensuring synthetic patient populations cover the full resilience spectrum. This self-improving property distinguishes the architecture from static pipeline designs where generation quality remains fixed post-deployment.

Reproducibility is ensured through fixed random seeds ($torch.manual_{\text{seed}(42)}$), deterministic CUDA operations (cudnn.deterministic=True), documented hyperparameters in configuration files, model checkpointing at regular intervals, and comprehensive logging of all training metrics. The complete pipeline executes via a single command (python run_pipeline.py) with expected outputs verified against reference benchmarks. All model artifacts are stored in standardized formats (.pth for PyTorch models, .npz for datasets, .pkl for scikit-learn models, .keras for TensorFlow models) with version-controlled code and Docker containerization for environment consistency.

VI. CONCLUSION

This paper presented Manō, a closed-loop generative AI framework for privacy-preserving mental health support. The four-component architecture demonstrates that clinically plausible AI systems can be developed under a strict zero-real-data paradigm while operating on consumer-grade hardware. The two-stage CTGAN+TimeGAN pipeline generates synthetic patients with 87.49% statistical fidelity and 96% temporal coherence. The hybrid LSTM achieves 96.2% classification accuracy with 14ms inference latency. The Seq2Seq world model enables RL-based treatment optimization converging within 5,000 episodes to clinically

interpretable policies. The GMM clustering system provides dynamic community formation with evidence-based activity recommendation.

The closed-loop architecture, where downstream performance metrics feed back to refine upstream generation, represents a self-improving system that can enhance synthetic data quality without exposing real patient information. The event-driven integration via Kafka enables real-time risk-aware conversational support, proactive intervention through trajectory forecasting, and dynamic community adaptation. All components maintain strict privacy guarantees by construction, with zero real patient data ingested, processed, or stored at any point in the pipeline.

Future work will focus on federated learning integration for real-data validation, uncertainty quantification via Bayesian methods, extended prediction horizons, adverse effect modeling, and multi-site clinical evaluation. The framework provides a foundation for accessible, privacy-preserving mental health AI that can scale to serve diverse populations without compromising individual privacy.

REFERENCES

[1] World Health Organization, "Mental Health Atlas 2020," WHO, Geneva, Switzerland, 2021.

[2] G. M. Weber et al., "Federated queries of clinical data repositories: the sum of the parts does not equal the whole," J. Am. Med. Inform. Assoc., vol. 20, pp. e155–e163, 2013.

[3] A. Rajkomar, J. Dean, and I. Kohane, "Machine learning in medicine," N. Engl. J. Med., vol. 380, no. 14, pp. 1347–1358, 2019.

[4] I. Goodfellow et al., "Generative adversarial nets," in Proc. Advances Neural Inf. Process. Syst. (NeurIPS), 2014, pp. 2672–2680.

[5] C. Esteban, S. L. Hyland, and G. Rätsch, "Real-valued (medical) time series generation with recurrent conditional GANs," arXiv:1706.02633, 2017.

[6] E. Choi et al., "Generating multi-label discrete patient records using generative adversarial networks," in Proc. Mach. Learn. Healthcare (MLHC), 2017, pp. 286–305.

[7] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in Proc. NeurIPS, 2019, pp. 7335–7345.

[8] J. Yoon, D. Jarrett, and M. van der Schaar, "Time-series generative adversarial networks," in Proc. NeurIPS, 2019, pp. 5508–5518.

[9] S. Dash, A. Yale, I. Guyon, and K. P. Bennett, "Medical time-series data generation using generative adversarial networks," in Proc. AIME, 2020, pp. 382–391.

[10] R. Donahue et al., "Synthesizing programmatically structured data for time-series tasks," in Proc. ICLR Workshop, 2019.

[11] A. G. Reece and C. M. Danforth, "Instagram photos reveal predictive markers of depression," EPJ Data Sci., vol. 6, no. 1, pp. 1–12, 2017.

[12] T. Shen et al., "Detecting risk of suicide attempts among Chinese medical college students using a machine learning algorithm," J. Affect. Disorders, vol. 273, pp. 18–23, 2020.

[13] Z. Che, S. Purushotham, K. Cho, D. Sontag, and Y. Liu, "Recurrent neural networks for multivariate time series with missing values," Sci. Rep., vol. 8, no. 1, pp. 1–12, 2018.

[14] E. Choi, M. T. Bahadori, J. Sun, J. Kulas, A. Schuetz, and W. Stewart, "RETAIN: An interpretable predictive model for healthcare using reverse time attention mechanism," in Proc. NeurIPS, 2016, pp. 3504–3512.

[15] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," in Proc. ICLR, 2015.

[16] O. Gottesman et al., "Guidelines for reinforcement learning in healthcare," Nat. Med., vol. 25, no. 1, pp. 16–18, 2019.

[17] L. Wang, W. Zhang, X. He, and H. Zha, "Supervised reinforcement learning with recurrent neural network for dynamic treatment recommendation," in Proc. ACM SIGKDD, 2018, pp. 2447–2456.

[18] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv:1707.06347, 2017.

[19] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," in Proc. NeurIPS Workshop, 2015.

[20] A. S. Fitzpatrick, K. K. Darcy, and M. Vierhile, "Delivering cognitive behavior therapy to young adults with symptoms of depression via a fully automated conversational agent (Woebot)," JMIR Ment. Health, vol. 4, no. 2, p. e19, 2017.

[21] H. Rashkin, E. M. Smith, M. Li, and Y.-L. Boureau, "Towards empathetic open-domain conversation models: A new benchmark and dataset," in Proc. ACL, 2019, pp. 5370–5381.

[22] S. Roller et al., "Recipes for building an open-domain chatbot," in Proc. EACL, 2021, pp. 300–325.

[23] Z. Lin, A. Madotto, J. Shin, P. Xu, and P. Fung, "MoEL: Mixture of empathetic listeners," in Proc. EMNLP, 2019, pp. 121–132.

[24] N. Papernot et al., "Semi-supervised knowledge transfer for deep learning from private training data," in Proc. ICLR, 2017.

[25] M. Abadi et al., "Deep learning with differential privacy," in Proc. ACM CCS, 2016, pp. 308–318.

[26] D. A. Reynolds, "Gaussian mixture models," in Encyclopedia of Biometrics, S. Z. Li and A. Jain, Eds. Boston, MA: Springer, 2009, pp. 659–663.

[27] A. K. Jain, "Data clustering: 50 years beyond k-means," Pattern Recognit. Lett., vol. 31, no. 8, pp. 651–666, 2010.

[28] X. Su and T. M. Khoshgoftaar, "A survey of collaborative filtering techniques," Adv. Artif. Intell., vol. 2009, pp. 1–19, 2009.

[29] P. Lops, M. de Gemmis, and G. Semeraro, "Content-based recommender systems: State of the art and trends," in Recommender Systems Handbook, F. Ricci et al., Eds. Springer, 2011, pp. 73–105.

[30] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Comput., vol. 9, no. 8, pp. 1735–1780, 1997.

[31] K. Cho et al., "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in Proc. EMNLP, 2014, pp. 1724–1734.

[32] A. Vaswani et al., "Attention is all you need," in Proc. NeurIPS, 2017, pp. 5998–6008.

[33] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171–4186.

[34] T. Brown et al., "Language models are few-shot learners," in Proc. NeurIPS, 2020, pp. 1877–1901.

[35] A. Radford et al., "Language models are unsupervised multitask learners," OpenAI Blog, vol. 1, no. 8, p. 9, 2019.

[36] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," in Proc. NeurIPS, 2014, pp. 3104–3112.

[37] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in Proc. ICLR, 2015.

[38] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in Proc. ICLR, 2019.

[39] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in Proc. ICML, 2015, pp. 448–456.

[40] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," J. Mach. Learn. Res., vol. 15, no. 1, pp. 1929–1958, 2014.

[41] V. Mnih et al., "Asynchronous methods for deep reinforcement learning," in Proc. ICML, 2016, pp. 1928–1937.

[42] J. Schulman, S. Levine, P. Abbeel, M. Jordan, and P. Moritz, "Trust region policy optimization," in Proc. ICML, 2015, pp. 1889–1897.

[43] V. Mnih et al., "Human-level control through deep reinforcement learning," Nature, vol. 518, no. 7540, pp. 529–533, 2015.

[44] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA: MIT Press, 2018.

[45] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in Proc. ICLR, 2016.

[46] P. Micikevicius et al., "Mixed precision training," in Proc. ICLR, 2018.

[47] T. Chen, B. Xu, C. Zhang, and C. Guestrin, "Training deep nets with sublinear memory cost," arXiv:1604.06174, 2016.

[48] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," J. R. Stat. Soc. B, vol. 39, no. 1, pp. 1–38, 1977.

[49] G. Schwarz, "Estimating the dimension of a model," Ann. Statist., vol. 6, no. 2, pp. 461–464, 1978.

[50] H. Akaike, "A new look at the statistical model identification," IEEE Trans. Autom. Control, vol. 19, no. 6, pp. 716–723, 1974.

[51] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," J. Comput. Appl. Math., vol. 20, pp. 53–65, 1987.

[52] D. L. Davies and D. W. Bouldin, "A cluster separation measure," IEEE Trans. Pattern Anal. Mach. Intell., vol. 1, no. 2, pp. 224–227, 1979.

[53] J. Lin, "Divergence measures based on the Shannon entropy," IEEE Trans. Inf. Theory, vol. 37, no. 1, pp. 145–151, 1991.

[54] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in Proc. NeurIPS, 2016, pp. 2234–2242.

[55] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in Proc. ICML, 2017, pp. 214–223.

[56] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. C. Courville, "Improved training of Wasserstein GANs," in Proc. NeurIPS, 2017, pp. 5767–5777.

[57] A. Brock, J. Donahue, and K. Simonyan, "Large scale GAN training for high fidelity natural image synthesis," in Proc. ICLR, 2019.

[58] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in Proc. ICLR, 2014.

[59] L. Xie, K. Lin, S. Wang, F. Wang, and J. Zhou, "Differentially private generative adversarial network," arXiv:1802.06739, 2018.

[60] B. K. Beaulieu-Jones et al., "Privacy-preserving generative deep neural networks support clinical data sharing," Circ. Cardiovasc. Qual. Outcomes, vol. 12, no. 7, p. e005122, 2019.

[61] R. Kessler et al., "Developing a practical suicide risk prediction model for targeting high-risk patients in the Veterans Health Administration," Int. J. Methods Psychiatr. Res., vol. 26, no. 3, p. e1575, 2017.

[62] M. De Choudhury, M. Gamon, S. Counts, and E. Horvitz, "Predicting depression via social media," in Proc. AAAI ICWSM, 2013.

[63] G. Coppersmith, M. Dredze, and C. Harman, "Quantifying mental health signals in Twitter," in Proc. ACL Workshop Comput. Linguist. Clin. Psychol., 2014, pp. 51–60.

[64] A. G. Reece et al., "Forecasting the onset and course of mental illness with Twitter data," Sci. Rep., vol. 7, no. 1, pp. 1–11, 2017.

[65] Z. C. Lipton, D. C. Kale, C. Elkan, and R. Wetzel, "Learning to diagnose with LSTM recurrent neural networks," in Proc. ICLR, 2016.

[66] E. Choi, A. Schuetz, W. F. Stewart, and J. Sun, "Using recurrent neural network models for early detection of heart failure onset," J. Am. Med. Inform. Assoc., vol. 24, no. 2, pp. 361–370, 2017.

[67] S. Razavi et al., "Clinical time series prediction with attention-based bidirectional LSTM," in Proc. KDD Workshop, 2019.

[68] L. Ouyang et al., "Training language models to follow instructions with human feedback," in Proc. NeurIPS, 2022, pp. 27730–27744.

[69] R. Anil et al., "PaLM 2 technical report," arXiv:2305.10403, 2023.

[70] H. Touvron et al., "LLaMA: Open and efficient foundation language models," arXiv:2302.13971, 2023.

[71] B. McMahan et al., "Communication-efficient learning of deep networks from decentralized data," in Proc. AISTATS, 2017, pp. 1273–1282.

[72] T. Li et al., "Federated learning: Challenges, methods, and future directions," IEEE Signal Process. Mag., vol. 37, no. 3, pp. 50–60, 2020.

[73] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in Proc. ICML, 2016, pp. 1050–1059.

[74] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?," in Proc. NeurIPS, 2017, pp. 5574–5584.

[75] J. Kramer, N. Rubin, S. Cotten, and T. Nolen, "Online and app-based cognitive behavioral therapy for insomnia: A systematic review and meta-analysis," Sleep Med. Rev., vol. 50, p. 101256, 2020.

[76] D. D. Burns, Feeling Good: The New Mood Therapy, rev. ed. New York: Avon Books, 1999.

[77] W. R. Miller and S. Rollnick, Motivational Interviewing: Helping People Change, 3rd ed. New York: Guilford Press, 2013.

[78] M. E. P. Seligman, Flourish: A Visionary New Understanding of Happiness and Well-being. New York: Free Press, 2011.

[79] C. Dwork, "Differential privacy: A survey of results," in Proc. Theory and Applications of Models of Computation (TAMC), 2008, pp. 1–19.

[80] J. Lu, D. Liu, and D. Diemert, "Machine learning for online recommendation systems," in Proc. WWW Tutorial, 2019.

[81] G. Linden, B. Smith, and J. York, "Amazon.com recommendations: Item-to-item collaborative filtering," IEEE Internet Comput., vol. 7, no. 1, pp. 76–80, 2003.

[82] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," Computer, vol. 42, no. 8, pp. 30–37, 2009.

[83] M. Kwon et al., "The smartphone addiction scale: Development and validation of a short version for adolescents," PLoS One, vol. 8, no. 12, p. e83558, 2013.

[84] J. B. Williams, "A structured interview guide for the Hamilton Depression Rating Scale," Arch. Gen. Psychiatry, vol. 45, no. 8, pp. 742–747, 1988.

[85] S. H. Lovibond and P. F. Lovibond, Manual for the Depression Anxiety Stress Scales, 2nd ed. Sydney: Psychology Foundation, 1995.

[86] A. Patel, S. S. Prabhu, and R. K. Muley, "Deep learning for mental health outcome prediction: A systematic review," Artif. Intell. Med., vol. 127, p. 102278, 2022.

[87] T. Tran, T. Pham, G. Carneiro, L. Palmer, and I. Reid, "Learning vector representation of medical objects via EMR-driven nonnegative restricted Boltzmann machines," J. Biomed. Inform., vol. 54, pp. 96–105, 2015.

[88] N. Jaques et al., "Predicting students' happiness from physiology, phone, mobility, and behavioral data," in Proc. ACII, 2015, pp. 222–228.

[89] F. Chikersal et al., "Detecting depression and predicting its onset using longitudinal symptoms captured by passive sensing," ACM Trans. Comput.-Hum. Interact., vol. 28, no. 1, pp. 1–41, 2021.

[90] J. W. Pennebaker, R. L. Boyd, K. Jordan, and K. Blackburn, "The development and psychometric properties of LIWC2015," Univ. Texas at Austin, 2015.